

BIS REFERENCE HANDBOOK: BUILDING MATERIALS & CIVIL ENGINEERING

STANDARDS COMPENDIUM FOR CEMENT, CONCRETE, TMT STEEL & NBC 2016

This handbook summarizes essential standards formulated by the Civil Engineering Division (CED) governing construction safety and structural integrity across India.

1. CEMENT STANDARDS SPECIFICATIONS

- **IS 269 : 2015:** Ordinary Portland Cement (OPC 33, 43, and 53 grades). Mandatory 28-day compressive strength of 53 MPa for OPC 53.
- **IS 1489 (Part 1 & 2):** Portland Pozzolana Cement (PPC - Fly Ash based and Calcined Clay based). High resistance to chemical/sulfate attacks.
- **IS 455 : 2015:** Portland Slag Cement (PSC) for marine and coastal construction.

2. PLAIN AND REINFORCED CONCRETE (IS 456 : 2000)

The foundational code of practice for structural RCC design in India. Specifies minimum cement content, maximum water-cement ratio, exposure conditions (Mild, Moderate, Severe, Very Severe, Extreme), and limit state design methodology.